

DATA VORTEX

FINAL EXPLORATORY DATA ANALYSIS REPORT

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Round 1 — Phase 1

Theme: Rebuilding the Social Engine

Scope	Verified value
Raw post records	12,360
Unique posts after de-duplication	12,000
User profiles	1,500
Post-user join coverage	100.0%

1. Objective

Round 1, Phase 1 requires restoring the Social Engine intake pipeline from a deliberately corrupted social-media dataset. This final EDA report combines the statistical and data-quality findings from the supplied EDA report with the supplied DataVortexEDAR1.ipynb workflow. The analysis covers recovery, missing-value handling, anomaly identification, data standardisation, descriptive statistics, temporal behaviour, platform differences, user behaviour, and cross-dataset relationships.

The report does not fabricate missing platform, text, or likes values. Transformations are limited to reversible formatting recovery, deduplication, timestamp standardisation, and conversion of logically impossible negative likes into missing values.

2. Dataset Structure

Dataset	Source	Rows	Columns	Grain
Posts	db2.csv / df2	12,360	8	One row per post record
Users	db1.csv / df1	1,500	5	One row per user

Post fields are post_id, user_id, platform, text_content, timestamp, likes, shares, and comments. User fields are user_id, location, language, account_created, and follower_count.

After duplicate removal, 12,000 unique posts remain. All 12,000 posts join to a user profile, giving 100.0% join coverage. Every one of the 1,500 users has at least one post. Posts per user range from 1 to 22, with a mean of 8.00.

3. Data Quality and Recovery

3.1 Missing values

Posts field	Raw missing	Missing %	Decision
post_id	0	0.00%	Keep
user_id	0	0.00%	Keep
platform	1,846	14.94%	Keep missing / Unknown
text_content	1,746	14.13%	Keep missing
timestamp	0	0.00%	Parse mixed formats
likes	1,858	15.03%	Keep missing
shares	0	0.00%	Keep
comments	0	0.00%	Keep

The corruption is concentrated in the post table: platform has 1,846 missing values (14.94%), text has 1,746 (14.13%), and likes has 1,858 (15.03%). The user table is structurally complete, with zero missing values across its five fields.

3.2 Duplicates and invalid values

There are 360 extra rows caused by duplicated post IDs, leaving 12,000 unique post IDs from 12,360 raw records. These duplicates represent 2.91% of the raw post table. The duplicate groups contain no conflicting versions; the 360 exact duplicate rows are repeated copies. The appropriate action is therefore to drop duplicate post_id records and keep the first occurrence.

There are 525 negative likes in the raw post data (4.25%). Likes cannot logically be negative, so negative values are converted to missing rather than replaced with zero, a mean, a median, or another guessed value. There are no negative shares and no negative comments.

3.3 Text and timestamp recovery

Recovery issue	Observed	Action
Mixed timestamp formats	4,950 ISO; 3,788 Unix seconds; 3,622 DD-MM-YYYY	Convert all to one datetime format
HTML entities	341 posts	HTML-unescape
HTML tags	663 posts	Strip tags; convert to spaces
UTF-8 / Latin-1 mojibake	316 posts	Reversible encoding repair where valid

The three timestamp encodings are representation differences rather than missing time. Standardisation is reversible and preserves the event date/time.

4. Cleaning Outcome

Measure	Before	After
Posts	12,360 rows	12,000 unique posts
Users	1,500	1,500
Usable non-negative likes	10,502 non-missing including negatives	9,677
Cleaned missing likes	1,858	2,323
Timestamp	3 mixed formats	YYYY-MM-DD HH:MM:SS; 0 unparsed
Join	—	12,000/12,000 matched (100%)

The cleaned pipeline produces 12,000 posts, 1,500 users, and a 12,000-row merged analytical table. The increase in missing likes from 1,858 to 2,323 reflects the addition of the unique-post negative likes that were correctly invalidated.

5. Statistical EDA — Post Engagement

Metric	N	Missing	Mean	Std. dev.	Min	Q1	Median	Q3	Max
Likes	9,677	2,323	2,493.51	1,438.29	0	1,240	2,502	3,724	5,000
Shares	12,000	0	1,007.17	575.07	0	510	1,018	1,501	2,000
Comments	12,000	0	504.35	288.68	0	253	503	755	1,000

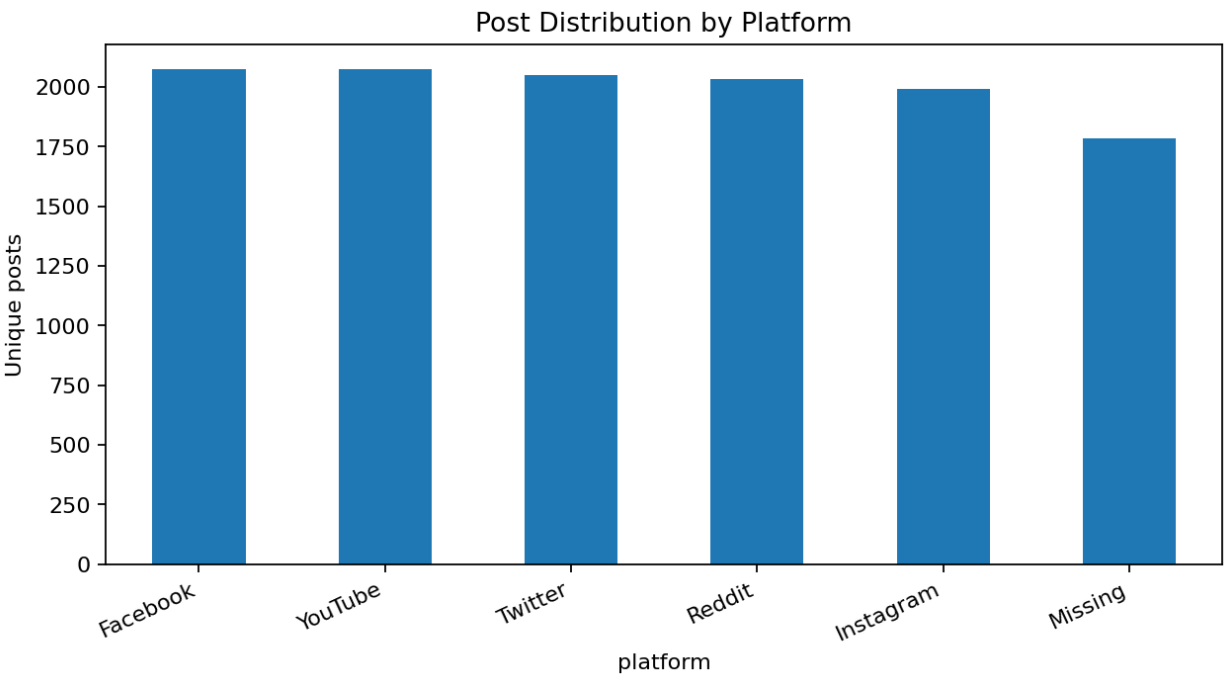
The cleaned engagement variables are tightly bounded: likes range from 0 to 5,000, shares from 0 to 2,000, and comments from 0 to 1,000. Mean likes are 2,493.51, mean shares are 1,007.17, and mean comments are 504.35. Corresponding medians are 2,502, 1,018, and 503.

Inference: the means lie close to the midpoints of the hard-looking ranges and skewness is near zero. The supplied report therefore characterises these variables as bounded, uniform-style draws rather than typical heavy-tailed social-media engagement. This is a substantive data-generation clue: the engagement columns appear unusually regular for organically generated social behaviour.

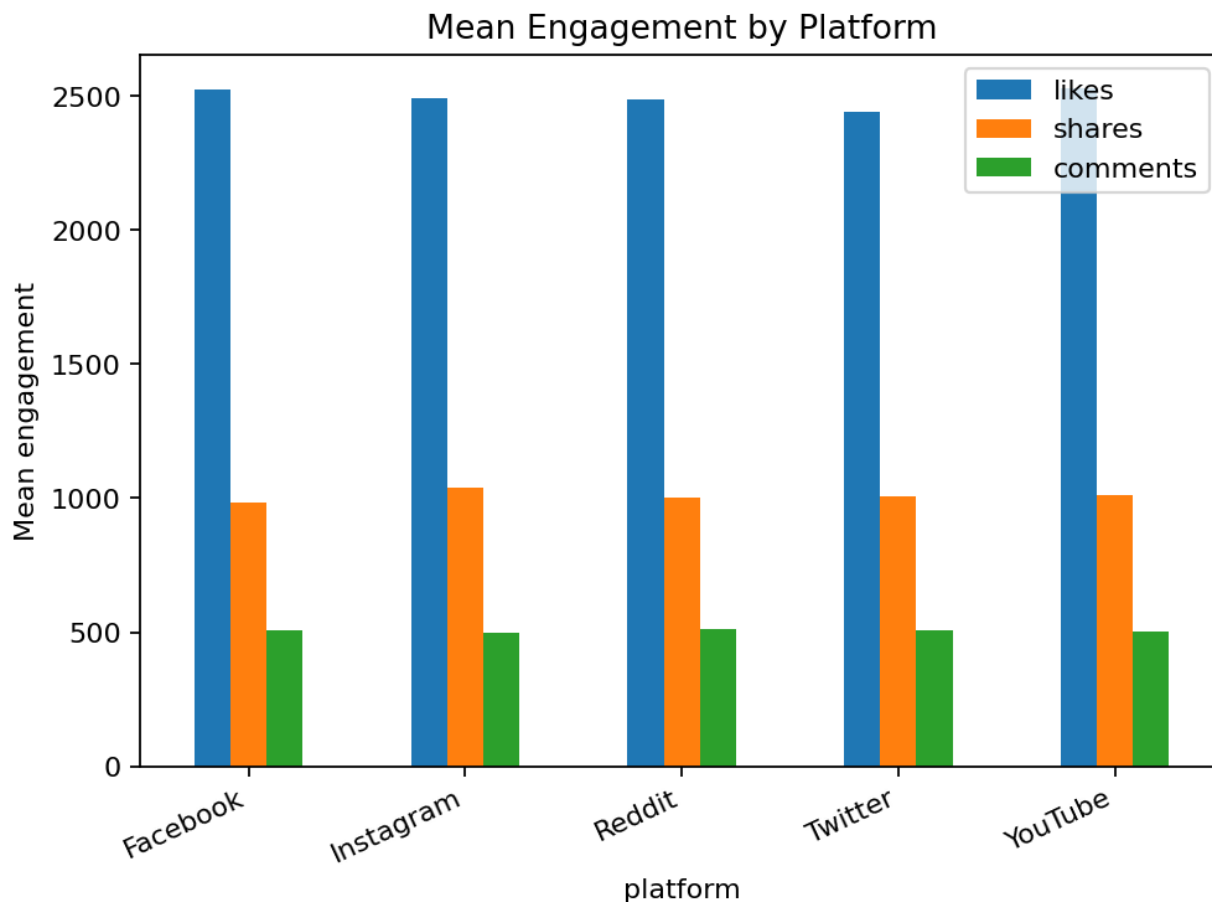
Zero engagement is rare: 1 valid post has zero likes, 5 have zero shares, and 13 have zero comments. Shares and comments remain usable even when likes are missing; dropping every row with any missing engagement value would unnecessarily discard about 19.4% of unique posts.

6. Platform Analysis

Platform	Raw	Unique	Share of unique	Mean likes	Mean shares	Mean comments
Facebook	2135	2074	17.28%	2526.55	984.17	506.94
YouTube	2136	2073	17.28%	2528.33	1011.83	504.38
Twitter	2119	2049	17.08%	2439.87	1005.39	506.13
Reddit	2086	2031	16.93%	2487.8	1002.23	511.18
Instagram	2038	1989	16.58%	2494.51	1040.84	499.8
Missing	1846	1784	14.87%	2480.82	998.61	496.54



Unique post distribution by platform.

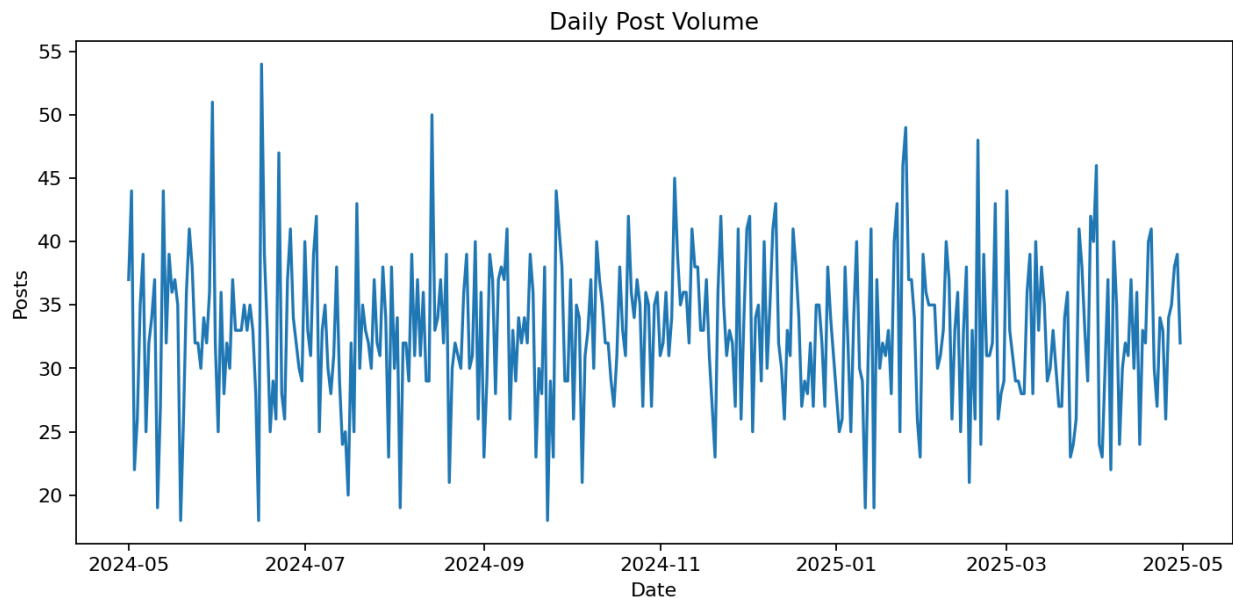


Mean engagement by recorded platform; missing platform is shown separately in the statistical table.

Named platforms are remarkably balanced, each contributing approximately 16.6%–17.3% of unique posts. The missing-platform category is itself large at 14.87%, nearly the size of any named platform.

Inference: mean likes span only 2,439.87–2,528.33 across the named platforms. This narrow range indicates that platform is not a strong splitter of engagement in this intake. Because platform is missing for a substantial fraction of records, platform-specific conclusions should still be treated cautiously.

7. Temporal Analysis



Daily post volume across the recovered one-year period.

Correct timestamp parsing gives complete coverage from 1 May 2024 through 30 April 2025: 365 distinct dates with no gaps. Daily volume has a mean of 32.88, median 33, minimum 18, and maximum 54 posts.

Monthly post volume:

Month	Posts
2024-05	1,038
2024-06	979
2024-07	996
2024-08	1,015
2024-09	974
2024-10	1,029
2024-11	1,025
2024-12	1,035
2025-01	1,007
2025-02	914
2025-03	1,013
2025-04	975

February 2025 is the lowest-volume month at 914 posts, which is consistent with February having fewer calendar days. The year is otherwise stable, with monthly totals remaining close to roughly one thousand posts.

Weekday activity and mean likes:

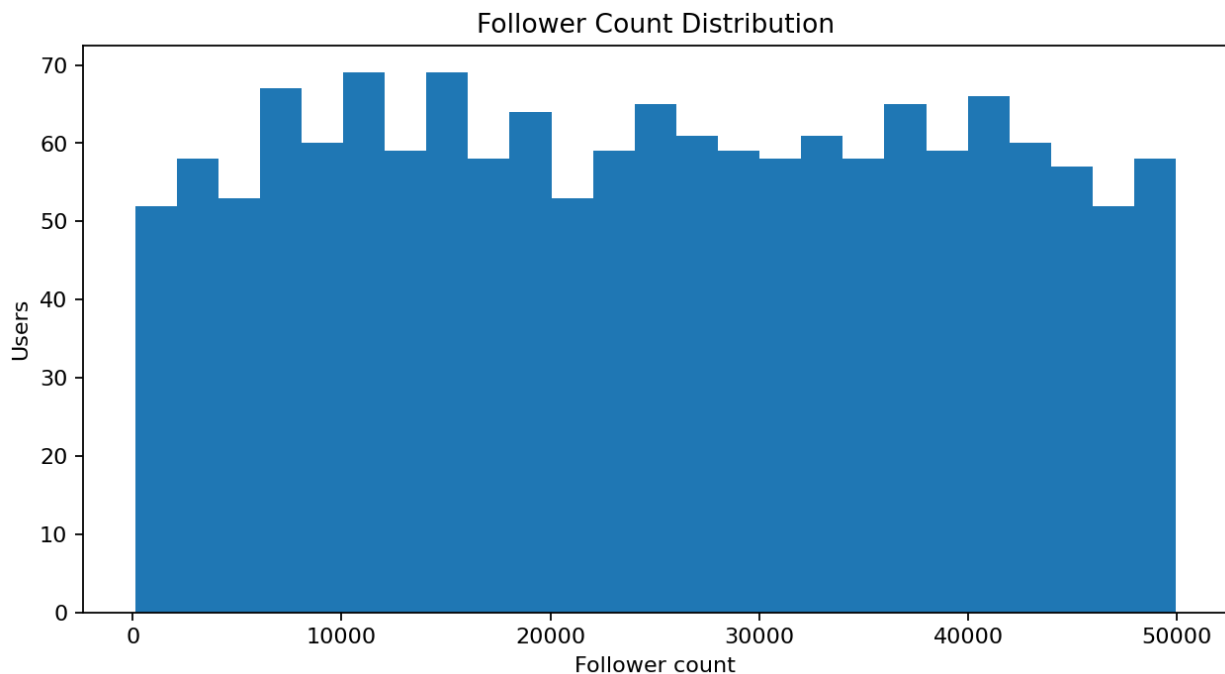
Weekday	Posts	Mean likes
Monday	1,720	2,487.69
Tuesday	1,677	2,468.99
Wednesday	1,771	2,510.90
Thursday	1,718	2,479.43
Friday	1,723	2,473.39
Saturday	1,675	2,502.05

Sunday	1,716	2,531.63
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Inference: weekday post counts remain close (1,675–1,771), and mean likes remain within a narrow 2,469–2,532 band. There is therefore no strong weekend/weekday engagement spike. The intake resembles a continuous, steady stream rather than bursty campaign data.

8. User-Level EDA

Follower statistic	Value
Count	1,500
Mean	24,964.02
Std. deviation	14,185.48
Minimum	109
25th percentile	12,771.75
Median	24,741.50
75th percentile	37,097.25
Maximum	49,944



Distribution of follower counts.

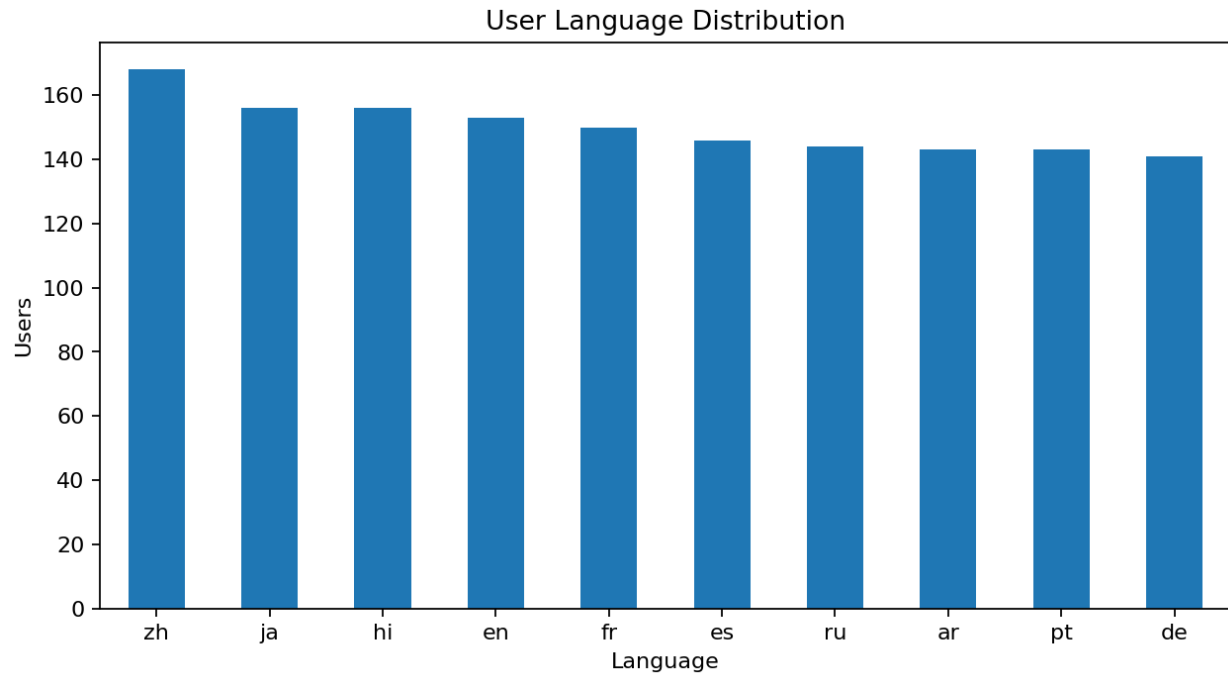
Follower count has mean 24,964.02, median 24,741.50, standard deviation 14,185.48, minimum 109, and maximum 49,944. The supplied report gives skewness of approximately 0.016.

Inference: follower count is nearly symmetric around 25,000 with a hard-looking upper bound near 50,000. This is unlike typical real social graphs, which are often strongly right-skewed. Combined with balanced languages and platforms, the user population appears synthetically balanced rather than organically concentrated.

8.1 Language and location

Language	Users
zh	168
ja	156
hi	156

en	153
fr	150
es	146
ru	144
ar	143
pt	143
de	141



There are 10 language codes, ranging from 141 to 168 users. The near-uniform language distribution reinforces the conclusion that the user panel is designed to be balanced. The supplied report identifies Shanghai and Barcelona as the largest individual cities, with 56 users each.

Country / territory	Users
USA	203
China	107
Germany	97
Italy	96
Japan	94
Brazil	93
Spain	92
India	92
UK	86
Canada	83
France	82
Australia	80
UAE	53
Singapore	49
South Africa	48
Mexico	40
Egypt	39
South Korea	36
Nigeria	30

The country distribution is broad rather than dominated by a single region: the USA has 203 users, followed by China (107), Germany (97), Italy (96), Japan (94), Brazil (93), Spain (92), and India (92).

9. User Posting Behaviour

Statistic	Posts per user
Users with ≥ 1 post	1,500/1,500
Mean	8.00
Std. deviation	2.85
Minimum	1
25th percentile	6
Median	8
75th percentile	10
Maximum	22

Every user contributes at least one post. Activity ranges from 1 to 22 posts per user, with mean 8.00 and standard deviation 2.85. The top 10 users contribute 1.47% of posts, the top 50 contribute 6.25%, and the top 100 contribute 11.61%.

Inference: activity is only mildly concentrated. The busiest user has 22 posts and the top 100 users account for only 11.61% of the corpus. This is not a power-user-dominated dataset and the exact 8.00 mean is consistent with a designed panel of 1,500 users and 12,000 posts.

10. Cross-Dataset Relationship Analysis

Variable	Likes	Shares	Comments	Follower count
likes	1.0000	-0.0029	0.0130	0.0084
shares	-0.0029	1.0000	0.0244	-0.0144
comments	0.0130	0.0244	1.0000	-0.0003
follower_count	0.0084	-0.0144	-0.0003	1.0000

Pearson correlations are essentially null: follower count versus likes is 0.0084, versus shares -0.0144, and versus comments -0.0003. Likes, shares, and comments are also nearly independent, with absolute correlations no larger than 0.024.

Inference: audience size does not track engagement in this dataset. This is supported not only by the correlation matrix but also by follower quartiles: mean likes are 2,462.17 in Q1, 2,503.50 in Q2, 2,500.71 in Q3, and 2,508.16 in Q4. The lowest-to-highest quartile difference is only about 46 likes on a roughly 2,500-like mean.

Follower quartile	Mean likes	Mean shares	Mean comments
Q1	2,470.58	1,023.14	504.52
Q2	2,496.03	1,003.76	505.52
Q3	2,501.94	997.90	500.31
Q4	2,506.29	1,003.64	507.11

Correlation is descriptive rather than causal. The result does not prove that followers can never affect engagement; it indicates that the supplied records contain no meaningful linear relationship and that the fields behave more like independently generated variables.

11. Text and Content Signals

Among non-missing raw text, 10,520 posts contain hashtags and 1,415 contain @-mentions. The supplied report identifies frequent hashtags such as Reviews (750), Fitness (749), BestValue (732), Eco (723), and SpecialOffer (723).

Inference: the unusually even hashtag frequencies, together with product/brand phrasing in sample posts, suggest generated review-style or promotional content rather than naturally evolving conversation. This is a dataset-generation clue and should be considered when evaluating models trained on the corpus.

12. Integrated Findings

The post table is the corrupted intake: 360 duplicate rows, mixed timestamp encodings, HTML/encoding residue, missing platform/text/likes, and 525 invalid negative likes.

De-duplication reduces the post table from 12,360 to 12,000 unique posts without losing conflicting versions.

No missing values were fabricated: missing platform, text, and likes remain missing, while negative likes are invalidated.

Timestamp corruption is representational rather than temporal: all three documented formats parse into a complete one-year timeline.

Engagement and follower counts are unusually range-bounded and near-symmetric, unlike typical organically generated social-media metrics.

Platforms, languages, locations, and posting frequency are unusually balanced, supporting the interpretation that the dataset is intentionally designed or synthetic.

Follower count has essentially zero linear relationship with likes, shares, and comments; engagement variables are also almost mutually uncorrelated.

Text is hashtag-heavy and promotional/review-oriented, with recoverable markup and encoding corruption.

to naturally distributed social-media data.

14. Reproducibility and Phase-1 Compliance

The supplied DataVortexEDAR1.ipynb loads both CSV datasets, profiles schema and missingness, detects duplicate post IDs and negative likes, parses the three timestamp formats, cleans reversible text corruption, deduplicates posts, joins users using user_id, and writes cleaned outputs. The supplied EDA report states that its numerical totals were recomputed from db1.csv and db2.csv using these same rules.

Requirement	Evidence in final report
Cleaned dataset	12,000 posts; 1,500 users; 12,000-row merge
EDA report	Quality, cleaning, statistics, temporal/platform/user analysis, correlations, inferences
Code notebook	DataVortexEDAR1.ipynb
Transformation justification	Duplicate, invalid-like, missing-value, timestamp, HTML/encoding decisions documented
Assumptions	post_id key, negative engagement invalid, no

	defensible imputation, stable user_id join
No fabrication	No platform/text/like imputation; negative likes converted to missing

Appendix A. Important Numerical Reference

Metric	Verified value
Raw posts	12,360
Unique posts	12,000
Duplicate post rows	360
Raw negative likes	525
Clean non-negative likes	9,677
Clean missing likes	2,323
Raw missing platform	1,846
Raw missing text	1,746
Raw missing likes	1,858
Users	1,500
Join coverage	100.0%
Mean likes (cleaned)	2,493.51
Mean shares	1,007.17
Mean comments	504.35
Mean followers	24,964.02
Median followers	24,741.50
Follower-likes Pearson r	0.0084
Time coverage	2024-05-01 to 2025-04-30
Distinct dates	365
Daily mean posts	33.86
Daily minimum	18
Daily maximum	59

Note: Where this final report presents cleaned statistics, they are based on unique posts after deduplication and with negative likes treated as missing. Raw missingness percentages are calculated against the original